

The Gravitational Register of Sun, Jupiter, Saturn, Earth

A personal monograph of the PLANETVORTEX closed gravifigure

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1. The model in one page

The Klein quartic $x^3y+y^3z+z^3x=0$ is the genus-3 surface whose automorphism group $PSL(2,7)$ of order 168 attains the Hurwitz bound $84(g-1)$. Its $\{7,3\}$ tessellation carries 24 heptagons — 12 antipodal pairs — and stage V2 of the bench carries the mass ladder of the chosen bodies in the vertex angles: $\alpha_i=2\pi/3-\sigma s_i$ with $s_i=\ln GM_i-\langle \ln GM \rangle$. The geometric-mean normalization makes $\sum s_i=0$ exactly, and that is precisely the condition that the decorated tessellation closes on the Gauss–Bonnet budget $\sum \alpha_i=8\pi=2\pi(2g-2)$: the Euler characteristic absorbs the mass ladder. The heavier the body, the wider its heptagon — gravity drawn as geometry, in exact closed form: $\cosh R_i=\cot(\alpha_i/2)\cot(\pi/7)$, $\cosh(\ell_i/2)=\cos(\pi/7)/\sin(\alpha_i/2)$, $\cosh r_i=\cos(\alpha_i/2)/\sin(\pi/7)$, $A_i=5\pi-7\alpha_i$.

2. The gravimetric register algebra of the chosen set

2.1
The
exact
registe
r
dimen
sions

body
GM, m³/s²
s

α , rad

R

ℓ

r

A

i, °

λ , rad

Sun

1.327124e+20

+6.9575

1.922675

0.915956

0.881879

0.779261

2.249242

—

—

Jupiter

1.266865e+17

+0.0033

2.094313

0.620843

0.566427

0.545417

1.047769

1.3044

0.022766

Saturn

3.793119e+16

-1.2026

2.124078

0.555220

0.501664

0.490318

0.839420

2.4860

0.043389

Earth

3.986004e+14

-5.7582
2.236515
0.138342
0.120360
0.124492
0.052360
0.0000
0.000000

$\lambda = \bar{R} \cdot i$ is the arc register of the spatial tilt against the ecliptic ($\bar{R} = 1$ AU); the inclination rows come from the committed J2000 register (JPL).

2.2 The budget closure of this set

The budget closure of this set: $\Sigma\alpha = 8.377580409573$ against $8\pi = 25.132741228718$ — residual $1.68e+01$ (**the figure does NOT close (check the normalization)**).

3. The spatial registers (the inclined sky)

The spatial registers of the bench: the tilt operators $T = R_z(\Omega)R_x(i) \in SO(3)$, the arc registers $\lambda_i = \dot{R}i_i$ and the mutual-inclination matrix are certified by stage V1 (protocol V1_inclined_registers.json); the 3D Newtonian run from the real J2000 sky carries the conservation of the energy and of the full angular-momentum vector.

4. The verification certificate of

this run

This monograph was generated together with a live verification run; the checks below were executed by the generator, not quoted.

#

check

result

recorded

1

$\Sigma\alpha = 8\pi$ (mpmath, 30 dps)

PASS

0.0

2

$\Sigma\alpha = 8\pi$ (float64)

PASS

7.11e-15

3

the hyperbolic Pythagoras (12 cells)

PASS

3.74e-16

4

the $SO(3)$ tilt orthogonality

PASS

2.22e-16

5

protocol P4_fano_algebra

PASS

committed

6

protocol X4_hyperbolic_figure

PASS

committed

7

protocol V2_klein_tiling

PASS

committed

5. The corpus and the reproduction commands

The formal corpus: Theorems A–C, Lemma D, Lemma E (the planar bench), Theorem F (the coset construction), Lemma F (the antipodal freeness), Theorem G (the budget closure), Lemma G (the arc register) — docs/theorems/.

Every plotted or quoted number of the bench is bound to a JSON protocol in results/protocols/ — committed before the run’s tolerance, refreshed after it.

Reproduce everything with the committed pipeline (from the mini-repository root):

```
make all-ladders  # P1..P7 + X1..X6 + V1..V2, JSON protocols
```

```
make test        # the pytest guard
```

```
make crosslang   # the C99 kernel vs Python
```

```
make figures600  # this gallery at 600 dpi
```

```
make monograph   # a sample monograph
```